

Clock-edge stepper

Sequential logic does not change on every input wiggle, it samples on a clock edge

Poke, edge, capture

- Starter lab: D flip-flop with a toggling clock
- Set D to one, apply poke, then advance to posedge clk, Q becomes one
- Set D to zero and take another posedge, Q returns to zero
- Enable registers only capture when EN is high
- Counters increment on edge after reset clears
- A pipeline needs two edges before the second stage sees your bit

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Clock-edge stepper

Drive a forever clock, poke data, and advance to the next posedge — the same step / edge / poke idea as the HDL workbench, focused on one register or counter.

Starter example: D flip-flop with a forever #5 clock. Poke D, then ⚡posedge clk (same idea as the HDL Simulator toolbar).

Load starter example

Challenges 0/22

Pick one

Capture a 1

With the D-FF lab, poke D=1 and advance to a posedge so Q becomes 1.

Start

Show hint

Next

Stop

Idle

Lab

D flip-flop
q <= d on every posedge clk. Poke d, then ⚡posedge.

Register + enable
q updates only when en=1 on posedge.

Signals now

clk0reg

d0reg

qzwire

Clock stepper starter

learn_digital — clock stepper

Workbook practice

- In the workbook track, sketch a D flip-flop timing diagram
- For an enable register with EN equals zero, say whether Q changes on posedge
- Draw two pipeline stages and label the one-cycle delay
- Name one pitfall: changing D and expecting Q to move before the clock edge

Pitfalls to watch

- Do not confuse level-sensitive latches with edge-triggered flops
- Reset and enable matter on the same edge as data
- And remember: the browser lab is literacy
- Real designs still need setup, hold

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one posedge capture wave
- When you are ready, take the short quiz, then continue to setup and hold